Question Bank for IAE 1

1. Explain OT (Operational Technology) and IT (Information Technology) networks, and differentiate between them.
2. Explain Li-Fi technology and its applications in the context of IoT.
3. Describe the physical design of IoT, including the various components and devices involved.
4. Define smart objects, and explain their characteristics and role in IoT.
5. Explain the logical design of IoT, including its architectural layers and protocols.
6. Describe Bluetooth Low Energy technology, its architecture, and its significance in IoT communication.
7. Explain Z-Wave technology, its working principle, and its applications in IoT.
8. Explain NFC and RFID technologies, and compare them in terms of their working, range, and applications.
9. Differentiate between Edge, Fog, and Cloud computing, and establish their hierarchical relationship.
10. With the help of a neat diagram, elaborate on the architecture of the IoTWF (IoT World Forum) reference model.
11. Explain the functional blocks of IoT and describe the role of each block.
12. Describe the various ZigBee devices and explain their operational modes and functions.
13. Differentiate between data analytics and network analytics in the context of IoT.
14. Differentiate between analytics and control applications in IoT systems.
15. Differentiate between data analytics and network analytics, and explain how data analytics contributes to business benefits.
16. Explain the core IoT functional stack, including its layers and the role of each layer.